

Alpha Chain: Final Closure

All conditions traced to canonical UBT sources

D. Jaros

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Abstract

We trace every condition in the UBT $\alpha^{-1} = 137.035999177549$ derivation to its canonical source in the repository. The derivation is [**L1 cond.**] with no free parameters. All conditions ultimately depend on the same bridge inputs as the T1_GR and T2_GAUGE papers. The complete chain is:

$$S[\Theta] \implies n^* = 137.035999177549 \implies \alpha_{\text{UBT}}^{-1} = n^*. \quad [\textbf{L1 cond.}]$$

1 Complete Condition Trace

Full alpha chain with condition sources.

Condition	Level	Source
<i>Mathematical fixed-point chain</i>		
$\Omega_\eta = \frac{1}{24} - \frac{1}{8\pi}$	[L0]	Eisenstein $G_2(i) = \pi$
$N_{\text{eff}} = 12 = 3 \times 4$	[L1]	su2.twist.neff12.tex
$Z_\psi[\tau] = \eta^{-12}$ (exact)	[L1]	SS BC + Gaussian path integral
$R_\psi = 1$	[L1]	modular fixed point
$r_{\text{L2}} = 3(1 + \Omega_\eta)$	[L1]	colour partition function
$\rho = e^{-C_Q/(2\pi n)}$ (exact)	[L1]	zero-mode equipartition
$n^* = 137.035999177549$	[L1]	fixed-point iteration
<i>Physical identification chain</i>		
$\Pi_{\text{EW}}\Theta \subset \mathcal{H}_{\text{odd,spinor}}$	[L1 cond.]	ew_odd_spinor_readout_theorem.tex
(i) $SU(2)_L = \text{left action on } \Theta$	[L1 cond.]	T2_GAUGE paper
(ii) $P_\psi = \gamma^5$, odd = left-handed	[L1 cond.]	Dirac eq. [[L1]] + ψ -parity [[L0]]
(iii) Odd winding = Grassmann	[L1]	fermionic_statistics_bridge.tex
$m_{0,\text{EW}}^2 = 0$ (H1)	[L1 cond.]	T-duality of $S[\Theta]$
$K_{\text{EW}} = e^2 > 0$ (H2)	[L0]	$ D_\psi\Theta ^2$ expansion
$\mu_{\text{EW}}^2 > 0$ (SSB)	[L1 cond.]	(i)–(iii) + H1 + H2
$\Theta_0 \sim (0, v/\sqrt{2}), Y = 1$	[L1 cond.]	theta0_nonzero_minimum_theorem.tex
$Q_{\text{EM}} = T_3 + Y/2$	[L1 cond.]	electroweak_em_projection_theorem.tex
$n = 4\pi/e^2 = \alpha^{-1}$	[L1 cond.]	em_modular_coupling_identification.tex
Result	[L1 cond.]	all above
$\alpha_{\text{UBT}}^{-1} = 137.035999177549$		

2 What the Conditions Depend On

All [L1 cond.] conditions ultimately trace to:

1. **T2_GAUGE bridge** [L1 cond.] — $SU(2)_L$ left action, hypercharge $Y = 1$. Same level as T2_GAUGE paper.
2. **Dirac equation in UBT** [L1] — $P_\psi = \gamma^5$ identification. Canonical result.
3. **T-duality of $S[\Theta]$** [L1 cond.] — $m_{0,\text{EW}}^2 = 0$. Same assumption as R-parity derivation.

4. **[L0] algebraic identities** — ψ -parity, $K_{\text{EW}} = e^2 > 0$, $Q_{\text{EM}} = T_3 + Y/2$. No assumptions beyond definitions.

No circular arguments. The mathematical chain ($n^* = 137.036$) uses only Z_ψ , N_{eff} , and $R_\psi = 1$ — all **[L1]** without any EW input. The physical identification uses T2_GAUGE and Dirac eq. as inputs, which are independent of α .

No free parameters. Every factor (4, 3, 12, Ω_η , C_Q) is derived, not chosen. The factor 4π in $n = 4\pi/e^2$ is the standard U(1) charge normalisation convention, not a fit.

3 Comparison with T1_GR and T2_GAUGE

Track	Result	Level	Key condition
T1_GR	$G_{\mu\nu}$ from Θ	[L1 cond.]	$\Theta \rightarrow g_{\mu\nu}$ bridge
T2_GAUGE	$SU(3) \times SU(2)_L \times U(1)_Y$	[L1 cond.]	involution structure
T3_ALPHA	$\alpha^{-1} = 137.035999177549$	[L1 cond.]	T2_GAUGE + Dirac + T-duality

Remark 3.1. The T3_ALPHA conditions are at the same level as T1_GR and T2_GAUGE. A reviewer who accepts the T2_GAUGE gauge derivation and the UBT Dirac equation should accept the full alpha chain.

4 Final Statement

Alpha derivation: COMPLETE within UBT bridge framework.

$$\alpha_{\text{UBT}}^{-1} = 137.035999177549 \dots$$

- Agreement with CODATA/NIST 2022 (137.035999177(21)): $+0.026\sigma$.
- No fitted parameter. No circular argument.
- All steps **[L1]** or **[L1 cond.]**.
- The **[L1 cond.]** conditions are the standard UBT bridge inputs shared with T1_GR and T2_GAUGE.
- External peer review status: not yet submitted. The claim is internal: “proven within the UBT bridge framework.”

Primary open tasks (not blocking the alpha result):

1. Physical scale M_{UBT} from $S[\Theta]$ (B5, needed for Weinberg angle).
2. Noether angular momentum computation for KK modes (B-fermionic, **[L2]**).
3. External peer review of T1_GR (next submission priority).

References

- [1] D. Jaros, `alpha_derivation_complete.tex`, 2026.
- [2] D. Jaros, `alpha_bridge_b1_closure.tex`, 2026.
- [3] D. Jaros, `em_modular_coupling_identification.tex`, 2026.
- [4] D. Jaros, `ew_odd_spinor_readout_theorem.tex`, 2026.
- [5] D. Jaros, `canonical/bridges/fermionic_statistics_bridge.tex`, 2026.
- [6] D. Jaros, `theta0_nonzero_minimum_theorem.tex`, 2026.
- [7] D. Jaros, `BRIDGE_CLOSURE_MATRIX.md`, 2026.